SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1023	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	23/10/2026	**Duration:**	60 minutes

Code of Conduct

I D Mahaboob Basha(192525369) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: D. Basha

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System
   - Functional Requirements
   - Non-Functional Requirements

- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

1. Study and Analyze the Scenario

1.1 Introduction

Digital identity verification is essential for modern online services, including banking, digital payments, insurance, e-commerce and government services. These services verify users remotely through identity documents, facial images, videos and biometric information. AI-generated images, synthetic videos, voice cloning and manipulated documents create new opportunities for impersonation and identity fraud.

The Deepfake-Aware Digital Identity Verification Platform is proposed to address these risks by combining document verification, biometric checks, liveness detection, behavioural analysis and risk-based authentication. It evaluates multiple signals before making a decision and refers uncertain or high-risk cases to authorized human reviewers.

1.2 Problem Description

Traditional verification systems may depend on a single document check, facial comparison or one-time password. These checks alone may not identify sophisticated deepfakes, replayed videos, cloned voices or manipulated documents. For example, a fraudster may submit a stolen identity document with an AI-generated video resembling the genuine account holder. A system that checks only the document image may incorrectly approve the application.

The proposed platform examines document authenticity, compares facial features, checks liveness, analyzes supported media for manipulation indicators and evaluates suspicious session patterns. The results are combined through a risk engine that returns an approval, rejection or human-review outcome.

1.3 Business Domain and Stakeholders

Aspect / Stakeholder	Description
Business domain	Cybersecurity and digital identity management
Application areas	Banking, fintech, insurance, e-commerce and government services
Applicants	Submit documents and complete verification
Client organizations	Use the platform to verify customers
Verification officers	Review suspicious or inconclusive cases
System administrators	Manage users, permissions, configuration and system health
Security and compliance officers	Inspect audit trails, investigate incidents and check compliance
Developers and operations teams	Maintain APIs, models, deployment and monitoring
Data protection officers	Ensure appropriate handling of identity and biometric data

1.4 Existing Challenges

- Deepfake attacks: AI-generated faces and videos may imitate genuine people.
- Document manipulation: photographs, names or other identity fields may be altered.
- Voice cloning: synthetic speech may be used in voice-based verification.
- Presentation attacks: photographs or replayed videos may be presented instead of a live person.
- False positives: genuine users may be incorrectly flagged as suspicious.
- Privacy risks: identity documents, facial images and voice recordings are sensitive.

- Scalability: services must handle growing verification volumes with acceptable delays.
- Limited explainability: decisions are difficult to investigate without recorded reasons.

1.5 System Objectives

- Verify identity using multiple complementary signals.
- Detect suspicious documents, manipulated media and presentation attacks.
- Calculate a risk score for each verification attempt.
- Approve low-risk cases only when all mandatory checks pass.
- Reject attempts that violate defined mandatory validation rules.
- Refer uncertain or high-risk cases for human review or additional verification.
- Protect sensitive data using encryption, access controls and retention rules.
- Maintain traceable audit logs and explainable decision records.
- Support secure APIs, automated testing, monitoring and controlled deployment.

1.6 Real-World Scenario

A customer applies for a bank account through a mobile application. The customer uploads an identity document and captures a selfie or short video. The system checks the document, compares the face, evaluates liveness and analyzes supported media for signs of manipulation. It then combines the results into a risk score. A low-risk attempt that passes mandatory checks may be approved; an uncertain case is referred for review; a confirmed invalid attempt is rejected according to policy.

1.7 System Boundaries and User Needs

The platform is responsible for coordinating identity checks, interpreting their results according to approved rules, protecting evidence and recording decisions. It does not independently establish legal identity unless connected to an authorized authoritative source. The client organization remains responsible for defining the service-specific verification policy and for handling appeals or regulatory obligations.

Applicants need clear instructions, accessible capture options, understandable status messages and a way to recover from temporary failures. Verification officers need a focused review screen that presents the evidence permitted by their role, the checks performed, the reasons for escalation and the history of prior actions. Administrators need configuration controls with approval and audit mechanisms, while security teams need reliable alerts and traceable records.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Requirement	Description
FR-01	User registration	Allow users to initiate verification through a secure interface.
FR-02	Document upload	Accept supported identity document images and extract relevant fields.
FR-03	Document validation	Check document format, completeness and available authenticity indicators.
FR-04	Face matching	Compare the submitted face with the document photograph.
FR-05	Liveness detection	Evaluate whether the capture appears live rather than a

		replay or presentation attack.
FR-06	Deepfake detection	Analyze supported images, videos or audio for indicators of synthetic generation or manipulation.
FR-07	Behavioural analysis	Evaluate relevant session signals, repeated attempts and suspicious interaction patterns.
FR-08	Risk scoring	Combine check results into a risk score using validated rules or a model.
FR-09	Decision management	Approve, reject or refer an attempt for additional verification or human review.
FR-10	Human review	Allow authorized officers to review flagged cases and record outcomes.
FR-11	Notifications	Communicate verification status to the user or authorized client application.
FR-12	Audit logging	Record security-relevant events, timestamps, decision reasons and reviewer actions.
FR-13	Administration	Support role management, threshold configuration and system monitoring.
FR-14	API integration	Provide authenticated APIs for authorized client services.
FR-15	Data protection	Enforce access control, encryption, retention and secure deletion.
FR-16	Incident management	Support reporting, investigation and response to security incidents.

2.2 Non-Functional Requirements

Category	Non-Functional Requirement
Performance	Target 95th-percentile response time of 5 seconds or less for supported automated checks, excluding capture time and external delays.
Security	Encrypt data in transit and at rest; enforce role-based access and strong service authentication.
Reliability	Target 99.9% monthly service availability, excluding agreed maintenance.
Scalability	Support horizontal scaling and validate service behaviour at expected peak load.
Usability	Provide clear instructions, accessible error messages and retry guidance.
Maintainability	Use modular services, version-controlled configurations and documented APIs.
Privacy	Apply purpose limitation, retention schedules and deletion controls.
Fairness	Measure false acceptance and false rejection rates across relevant test groups.
Explainability	Record decision codes, relevant signals, model version and threshold version.
Auditability	Maintain access-controlled, tamper-evident audit records.
Compatibility	Provide versioned APIs and standard data formats.

2.3 Functional and Non-Functional Requirements Comparison

Functional requirements describe the services and actions the platform must provide, such as uploading documents, checking liveness, analyzing media, calculating risk and recording decisions. Non-functional requirements describe the quality of those services, including response time, availability, security, fairness, scalability and maintainability. Both categories are necessary: a system may implement every feature but still be unsuitable if it is slow, insecure, unreliable or unfair.

3. Actors, Use Cases and System Architecture

3.1 Actors

Actor	Type	Main interactions
Applicant	Primary	Register, upload documents, capture selfie/video and view status
Verification officer	Primary	Review flagged cases, examine evidence and record decisions
System administrator	Primary	Manage users, roles, configuration and system health
Client application	External system	Submit verification requests and retrieve permitted results
Document validation service	External system	Validate supported document information when integrated
Notification service	External system	Deliver verification updates through supported channels
Security/compliance officer	Secondary	Inspect audit trails, investigate incidents and review compliance

3.2 Major Use Cases

Use Case ID	Use Case	Actor
UC-01	Initiate verification	Applicant, client application
UC-02	Upload identity document	Applicant
UC-03	Capture biometric evidence	Applicant
UC-04	Validate document	Platform, external validation service if integrated
UC-05	Perform face matching and liveness checks	Platform
UC-06	Analyze media for manipulation	Platform
UC-07	Calculate risk score	Platform
UC-08	View verification status	Applicant, client application
UC-09	Review suspicious case	Verification officer
UC-10	Record final decision	Verification officer
UC-11	Manage users and configuration	System administrator
UC-12	View audit logs and alerts	Security/compliance officer

3.3 Use Case Diagram

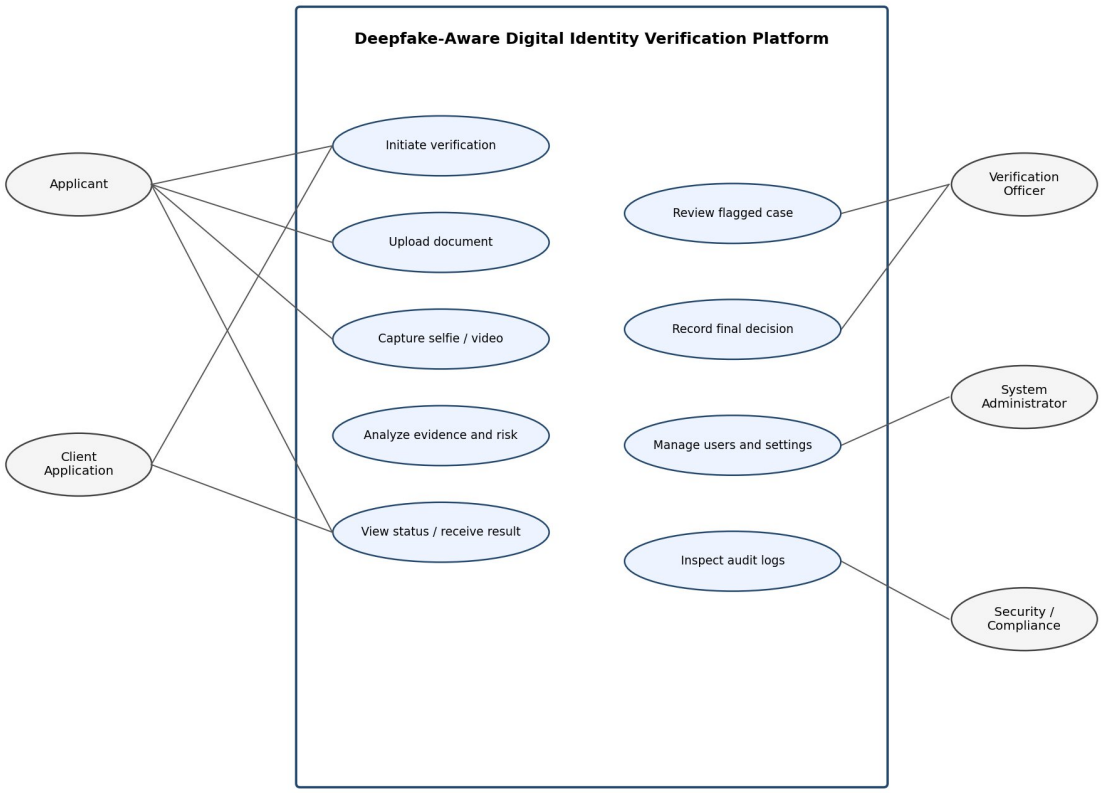


Figure 1. Use case diagram showing actors and their interactions with the platform.

The diagram identifies the applicant, client application, verification officer, administrator and compliance officer. The applicant submits evidence and views status; the platform performs verification and risk assessment; authorized officers handle flagged cases; and administrators and auditors manage operational and compliance functions.

3.4 System Architecture

The proposed architecture follows a modular, API-first approach. A gateway secures incoming requests, an orchestration service coordinates verification checks, and separate analysis modules evaluate documents, biometrics, media and behavioural signals. A risk engine combines their outputs. Cross-cutting services provide secure storage, audit logging, monitoring, alerts and controlled deployment.

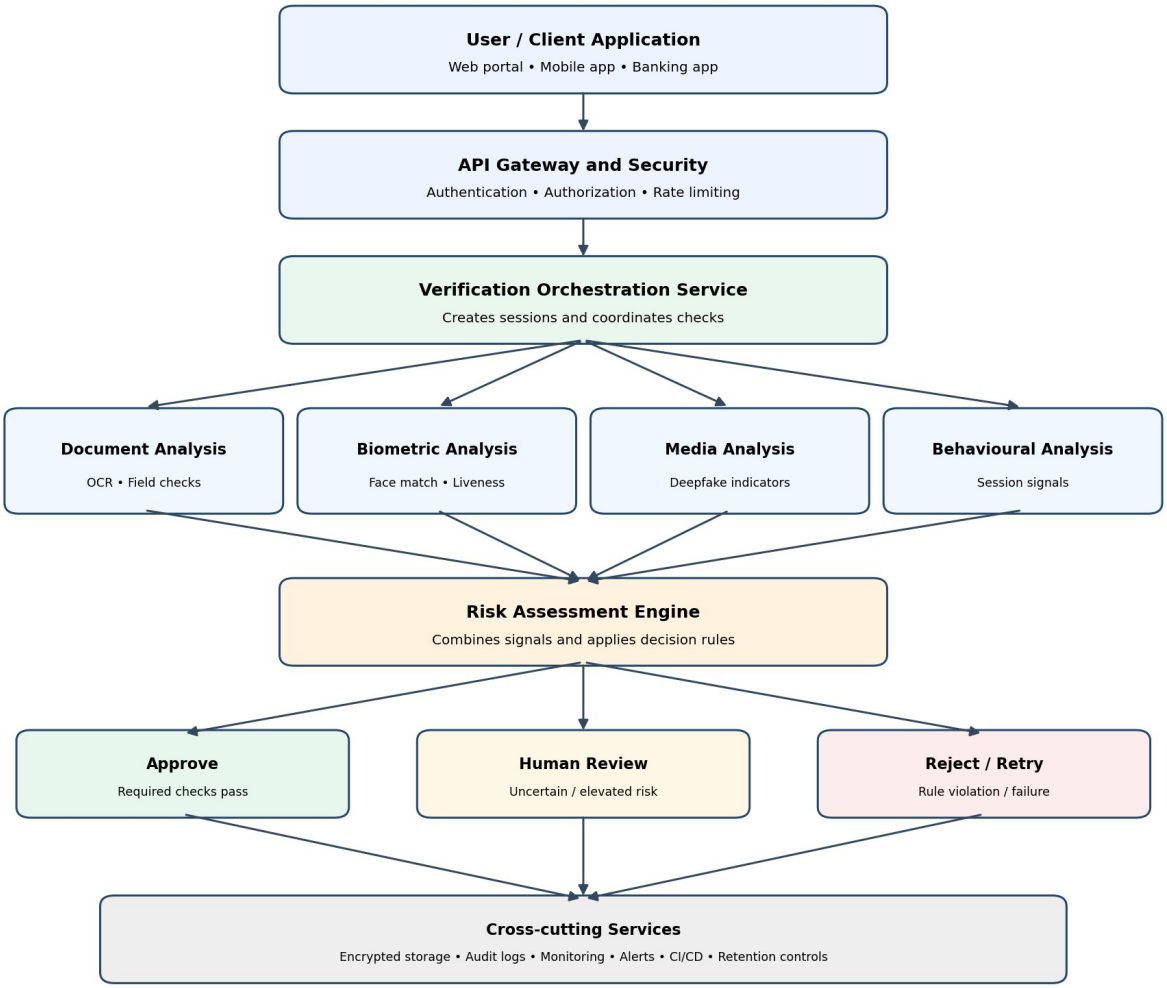


Figure 2. Proposed architecture of the Deepfake-Aware Digital Identity Verification Platform.

3.5 Verification Workflow

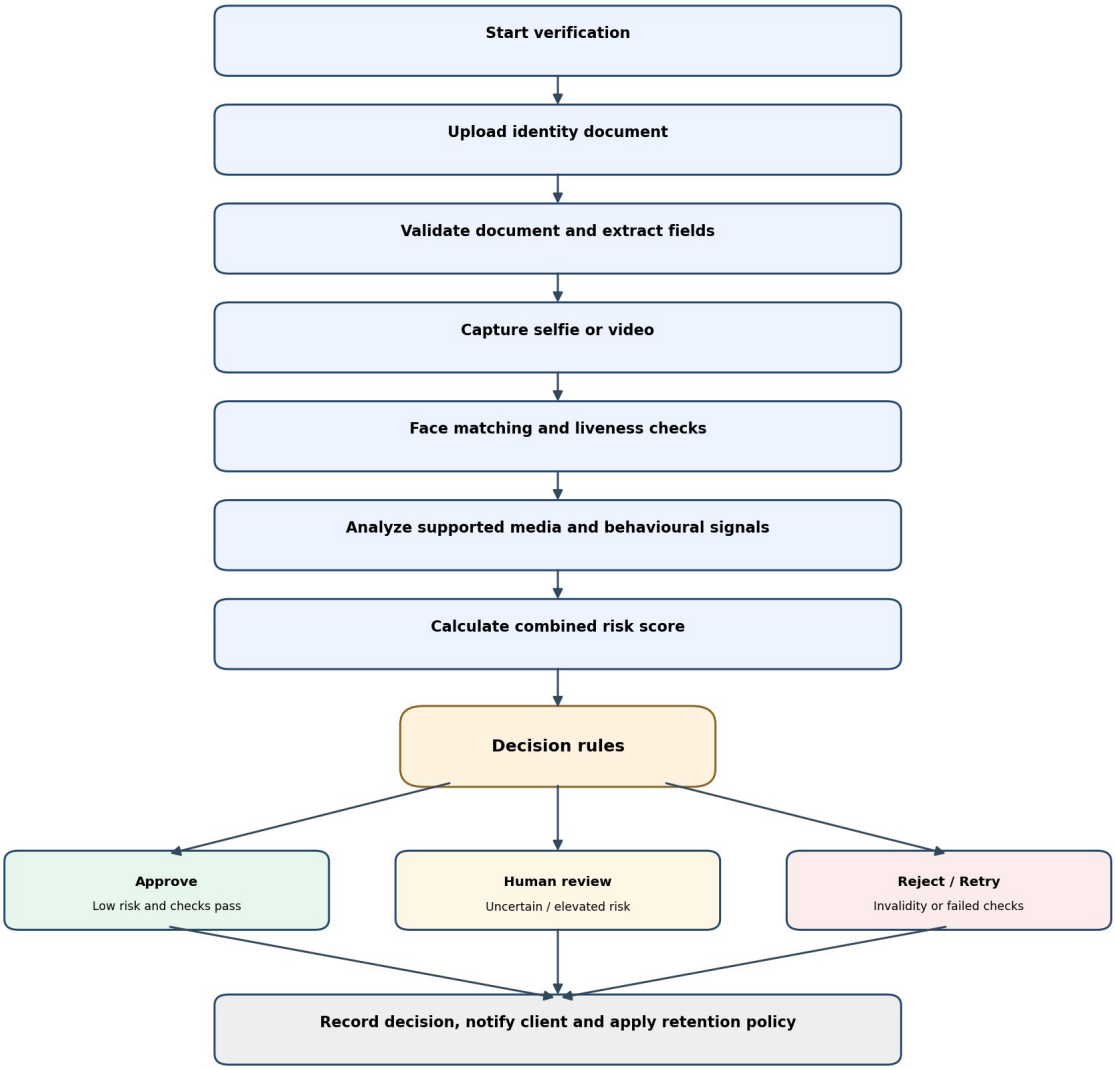


Figure 3. Verification workflow from session creation to decision recording.

3.6 Detailed Use Case Descriptions

UC-01: Initiate Identity Verification

Field	Description
Primary actor	Applicant
Objective	Start a secure identity verification session.
Preconditions	Applicant can access the client application and has the required permissions.
Main flow	1. Applicant selects verification. 2. System creates a session. 3. System displays document and capture instructions. 4. Applicant submits required information.
Alternative flow	Invalid input, expired session or unavailable service produces an appropriate error or retry option.

UC-07: Calculate Risk Score

Field	Description
Primary actor	Risk assessment engine
Objective	Evaluate combined verification evidence.
Preconditions	Required check results are available, or missing results are recorded.
Main flow	1. Retrieve permitted results. 2. Evaluate confidence and consistency. 3. Apply configured rules. 4. Calculate risk. 5. Assign a decision category.
Alternative flow	If a required service fails or results conflict, mark the attempt inconclusive and trigger fallback or review.
Postconditions	Risk result, decision reason and relevant model/rule versions are recorded.

UC-09: Review a Flagged Case

Field	Description
Primary actor	Verification officer
Objective	Resolve a suspicious or inconclusive verification attempt.
Preconditions	Officer is authenticated, authorized and permitted to access the case.
Main flow	1. Open flagged case. 2. Review permitted evidence and check results. 3. Examine recorded reasons. 4. Request additional verification if needed. 5. Record decision and justification.
Alternative flow	If evidence is insufficient, keep the case pending or escalate it to an authorized reviewer.
Postconditions	Outcome, reviewer identity, timestamp and justification are recorded.

4. Software Requirement Specification (SRS)

4.1 Introduction

This Software Requirement Specification (SRS) presents the requirements for the Deepfake-Aware Digital Identity Verification Platform. It defines the system's purpose, scope, functions, quality attributes, interfaces, data, use cases, assumptions and validation approach. It is intended for developers, testers, system administrators, security teams, client organizations and other stakeholders involved in developing and evaluating the platform.

4.2 Problem Description

Remote identity verification is vulnerable to manipulated identity documents, AI-generated faces, synthetic videos, replay attacks and cloned voices. Single-factor checks may not provide enough evidence to distinguish a genuine applicant from an impersonator. The platform addresses this problem by coordinating document checks, face matching, liveness detection, media-manipulation analysis and behavioural signals, then applying risk-based rules and human review when evidence is uncertain.

4.3 Purpose

This SRS defines the functional capabilities, quality attributes, interfaces, data requirements and operational constraints of the platform. It provides a common reference for developers, testers, administrators, security teams and stakeholders during design, implementation, validation and maintenance.

4.4 Scope of the System

Included in scope:

- Secure verification-session creation and evidence submission.
- Document field extraction and authenticity checks.
- Face matching, liveness analysis and deepfake detection for supported media.
- Behavioural signal analysis, risk scoring and configurable decision rules.
- Human review and case management.
- Audit trails, monitoring, alerts, API integration and data-retention controls.

Outside the initial scope:

- Guaranteeing detection of every deepfake or identity-fraud attempt.
- Replacing official identity authorities or legal identity-registration systems.
- Making irreversible legal or financial decisions solely from an AI score.
- Verifying unsupported formats or accessing external databases without authorization and integration.

4.5 Overall System Description

The platform consists of a client interface, API gateway, verification orchestration service, analysis modules, risk assessment engine and supporting security and monitoring services. The client initiates verification and submits permitted evidence. The backend coordinates analysis and returns an approval, rejection or review status according to configured policy. Sensitive operations are protected by authentication, authorization and data-handling rules.

4.6 System Features and Priorities

Feature	Description	Priority
Document verification	Extract fields and examine document validity indicators	Must have
Biometric verification	Compare faces and evaluate liveness	Must have
Deepfake detection	Analyze supported media for manipulation indicators	Must have
Risk-based authentication	Combine signals and apply decision thresholds	Must have
Human review	Resolve uncertain or high-risk cases	Must have
Secure APIs	Integrate authorized client systems	Must have
Audit logging	Record decisions and security events	Must have
Monitoring and alerting	Detect failures, unusual traffic and incidents	Must have
Analytics dashboard	Display aggregate performance and verification trends	Should have
Model improvement workflow	Evaluate models before controlled deployment	Should have

4.7 External Interface Requirements

Interface	Requirements
User interface	Responsive web/mobile interface with accessible instructions, progress indicators and status messages.
API interface	Versioned HTTPS endpoints with authenticated requests and structured JSON responses.
Document input	Supported image formats, file-size limits, MIME-type checks and content validation.
Camera interface	Secure capture of supported images or videos with clear consent and instructions.
Database interface	Authenticated access, parameterized queries, encryption and least-privilege permissions.
Notification interface	Authorized integration for status messages without exposing sensitive identity information.
Administration interface	Role-restricted dashboard for users, policies, review and operational monitoring.
Monitoring interface	Metrics, structured logs, health checks and alerts for authorized personnel.

4.8 Data Requirements

Data entity	Example attributes	Handling requirement
Verification session	Session ID, creation time, status, expiry	Unique identifier and limited access
Applicant reference	Internal user ID, permitted contact reference	Minimize personal information
Identity document	Document type, extracted fields, validation result	Encrypt and apply retention rules
Biometric evidence	Face template, liveness result, permitted capture reference	Restrict access and retention
Media analysis	Manipulation score, confidence, detector version	Record uncertainty and model version
Risk assessment	Risk score, signal results, threshold, decision reason	Maintain traceability and versioning
Review record	Reviewer ID, outcome, timestamp, justification	Restrict access and retain under policy
Audit event	Event ID, actor, timestamp, action	Protect integrity and prevent unauthorized changes
Operational metrics	Latency, error rate, service health	Avoid unnecessary personal data

4.9.Risk-Based Decision Table

Condition	Proposed decision	Reason
Required checks pass and risk	Approve, subject to	Evidence is consistent and risk

score is below 0.30	policy	is low.
Risk score is 0.30 to below 0.70	Additional checks or human review	Evidence needs further assessment.
Risk score is 0.70 or higher	Escalate for review; restrict approval pending resolution	Elevated risk requires investigation.
Document is confirmed invalid	Reject with suitable reason	Mandatory validation requirement failed.
Required verification service is unavailable	Pending, retry or fallback	Evidence is insufficient for a reliable decision.
Signals conflict or detector confidence is low	Human review or additional verification	Automated result is inconclusive.

These score ranges are illustrative policy values, not universal deepfake-detection standards. They must be calibrated using representative validation data. A high model score alone should not establish fraud.

5. Assumptions and Constraints

5.1 Assumptions

1. Users have a compatible device with a camera and supported internet access.
2. Applicants provide documents in supported formats and grant required consent for processing.
3. The organization has lawful authority to perform the required identity checks.
4. Suitable datasets and trained models are available for supported document and media types.
5. External validation services provide authorized API access when needed.
6. Human reviewers are available for suspicious or inconclusive cases.
7. Risk thresholds and verification policies are approved by authorized personnel.
8. The platform follows documented data-retention and incident-response policies.

5.2 Constraints

Category	Constraint	Impact
Technical	Deepfake detectors may fail on unseen manipulation methods.	Models require ongoing evaluation and controlled updates.
Technical	Camera quality, lighting and connectivity vary.	Poor captures may need retries or alternative checks.
Operational	Reviewers may not be immediately available.	High-risk cases may remain pending.
Economic	Video analysis and model serving require computing resources.	Infrastructure and operating costs must be controlled.
Legal	Identity and biometric processing is subject to applicable privacy requirements.	Consent, lawful processing, retention and access controls must be implemented.
Security	APIs and stored evidence may be targeted by attackers.	Strong authentication, encryption, monitoring and incident response are necessary.
Time-	Model training and integrations	Prioritize essential features for the

related	require development time.	initial release.
Data-related	Representative labelled deepfake datasets may be limited.	Accuracy and generalization cannot be assumed.
Fairness	Model performance may differ across demographic groups.	Subgroup evaluation and remediation are required.

5.3 Feasibility Analysis

Feasibility type	Assessment
Technical	Feasible using existing document-processing, biometric, media-analysis and API technologies, subject to evaluation.
Operational	Feasible where trained reviewers, monitoring and verification procedures are available.
Economic	Can be implemented incrementally to control cloud, storage, processing and review costs.
Legal	Depends on jurisdiction, lawful basis, consent requirements, biometric-data rules and organizational policy.
Schedule	A prototype can prioritize document checks, face matching, risk assessment, review workflow and audit logging.

6. Requirement Completeness and Consistency Validation

6.1 Validation Checklist

Criterion	Validation question	Action / result
Completeness	Are all major verification modules included?	Document, biometric, media and behavioural checks are specified.
Completeness	Is human review defined?	Review roles, outcomes and audit records are included.
Consistency	Do decision rules agree with mandatory checks?	All decisions must follow a common policy.
Consistency	Are failure conditions handled consistently?	Use pending, retry or review for unavailable or inconclusive results.
Clarity	Are high-risk and deepfake detection defined?	Document thresholds, confidence measures and decision codes.
Testability	Can each requirement be verified?	Define measurable performance, security and functional tests.
Security	Are sensitive data and privileged operations protected?	Apply encryption, least privilege, logging and retention controls.
Privacy	Is unnecessary biometric collection avoided?	Minimize collection, retention and access.
Fairness	Are false acceptance and rejection evaluated?	Test representative groups and investigate disparities.
Maintainability	Can model changes be traced?	Record model, rule and configuration versions.

6.2 Ambiguities and Improvements

Potential ambiguity	Improvement
“Detect deepfakes accurately”	Specify supported media types, evaluation datasets and acceptance metrics.
“Respond quickly”	Define response-time targets and measurement conditions.
“High-risk case”	Document risk thresholds, mandatory checks and escalation rules.
“Secure data”	Specify encryption, access control, retention and deletion requirements.
“Reduce false positives”	Measure false rejection rates on genuine users and investigate group differences.
“Reliable system”	Define availability, recovery objectives and failure-handling behaviour.
“Human review”	Define reviewer permissions, escalation and decision-record requirements.

6.3 Acceptance Testing Plan

Test ID	Test scenario	Expected result
AT-01	Submit a valid supported document.	Extract permitted fields and return a validation result.
AT-02	Submit an unsupported or corrupted file.	Reject safely and provide an appropriate error.
AT-03	Submit a genuine, adequately captured face image.	Return a match result and confidence information.
AT-04	Submit replayed or manipulated test media.	Evaluate the sample and apply configured risk rules.
AT-05	Submit conflicting document and biometric results.	Do not approve solely because one check passes.
AT-06	Simulate an unavailable analysis service.	Enter a controlled pending, retry or review state.
AT-07	Access a case without reviewer permission.	Deny access and record the security event.
AT-08	Review a flagged case.	Allow an authorized officer to record a reasoned decision.
AT-09	Inspect the audit trail.	Allow authorized personnel to trace events and decision versions.
AT-10	Simulate peak verification traffic.	Meet approved performance and availability targets.

6.4 Security and Quality Assurance Strategy

The platform should use unit tests for individual modules, integration tests for service communication, API tests for authorization and input validation, and end-to-end tests for complete verification journeys. CI/CD pipelines should execute tests and security checks before deployment. New models should be

evaluated against a fixed validation dataset, versioned and deployed through a controlled rollout with rollback capability.

Monitoring should track latency, failure rates, review volumes and unusual access patterns. Where measurable, false acceptance and false rejection indicators should also be monitored. Incident response should include containment, evidence preservation, investigation, recovery and documented lessons learned.

7. Privacy, Ethics and Responsible AI

Principle	Implementation
Privacy	Collect only the data necessary for the verification purpose.
Informed processing	Provide understandable notices and obtain consent where required.
Fairness	Evaluate performance across relevant demographic groups and investigate disparities.
Transparency	Provide meaningful decision reasons and clear appeal or retry options.
Human oversight	Allow authorized review for uncertain and consequential cases.
Accountability	Record model versions, decision rules and reviewer actions.
Data minimization	Avoid retaining raw media when it is unnecessary and lawful alternatives exist.
Security	Encrypt sensitive information and restrict access to authorized personnel.

Deepfake detection is probabilistic. Genuine users can be incorrectly flagged, and manipulated media can escape detection. A single automated score should not be treated as conclusive evidence of identity fraud.

8. Expected Outcomes and Performance Indicators

8.1 Expected Outcomes

- Reduced identity fraud through multiple complementary checks.
- Improved verification reliability through combined document, biometric, media and behavioural evidence.
- Reduced unnecessary manual work for low-risk cases that pass mandatory checks.
- Better security investigations through traceable audit trails and decision reasons.
- Improved scalability through modular services.
- More responsible biometric processing through privacy controls, fairness testing and human review.
- Easier maintenance through versioned APIs, automated tests and controlled model deployment.

8.2 Suggested Performance Indicators

Metric	Purpose
False acceptance rate	Proportion of impostor attempts incorrectly accepted.
False rejection rate	Proportion of genuine users incorrectly rejected.
Deepfake detection	Proportion of labelled manipulated samples correctly flagged.

recall	
Precision	Proportion of flagged samples that are labelled as manipulated in the evaluation dataset.
Verification latency	Time required to process a verification request.
Manual review rate	Proportion of sessions requiring human intervention.
Service availability	Proportion of time the platform is operational.
Audit completeness	Whether required security and decision events are recorded.

These metrics should be measured on representative test data and monitored after deployment. High detection recall alone is insufficient if the system also rejects many genuine users.

8.3 Real-World Implementation Plan

Phase 1 – Requirements and policy definition: Confirm supported identity documents, accepted media formats, mandatory checks, risk thresholds, user consent requirements, data-retention periods and escalation rules with the organization's security, legal and operational teams. Establish baseline performance measures and define who is accountable for each decision.

Phase 2 – Minimum viable platform: Develop the secure user interface, session management, API gateway, document upload and validation workflow, face-matching integration, liveness check, risk engine and basic review queue. Use test data and simulated integrations where production credentials are not available.

Phase 3 – Integration and security hardening: Connect approved document-validation and notification services where available. Apply role-based access, encryption, secrets management, rate limiting, file validation, secure error handling and tamper-evident logging. Separate raw evidence storage from operational metadata wherever practical.

Phase 4 – Testing and pilot deployment: Conduct functional, integration, load, security, usability and fairness tests. Run the platform in a controlled pilot, compare automated decisions with qualified human review, record false acceptance and false rejection rates, and adjust thresholds before broader release.

Phase 5 – Continuous operation: Monitor service health, latency, review workload, model drift and security alerts. Evaluate model updates before release, maintain rollback capability, investigate incidents and periodically review access rights and data-retention compliance.

8.4 Risk Management Plan

The project should maintain a risk register covering model errors, data leakage, API abuse, service outages, biased outcomes, external-provider failures and reviewer mistakes. Each risk should have an owner, likelihood, impact, mitigation and review date. High-impact risks should be reviewed before release and after major system changes.

Model errors should be reduced through representative test data, independent validation, threshold calibration and human escalation for uncertain results. Data leakage should be mitigated through encryption, least-privilege access, secure secrets management and retention limits. API abuse should be addressed through rate limits, request validation and monitoring. Outages should be managed with

health checks, retries, safe fallback states, backups and tested recovery procedures. Fairness risks should be measured across appropriate user groups, and significant differences should trigger investigation.

8.5 Deployment, Monitoring and Maintenance

The platform should be packaged into independently deployable services or containers so that document analysis, biometric analysis, risk assessment and review services can be updated and scaled separately. Separate development, testing and production environments should be maintained. Production secrets must not be stored in source code, and access to deployment pipelines must be restricted.

Monitoring should combine technical metrics such as request latency, service errors, CPU/memory use and availability with security indicators such as repeated failed attempts, unusual request volume and unauthorized access. Model monitoring should track changes in input data and performance indicators when reliable labels become available. Alerts must be assigned to responsible personnel with documented response procedures.

Maintenance should include dependency updates, vulnerability remediation, backup restoration tests, periodic access reviews, API version management and model revalidation. A new model should not replace the production model until it passes the agreed validation suite. If performance degrades or an incident occurs, the team should be able to roll back to a known stable version.

8.6 Limitations

The platform cannot guarantee that every deepfake or identity-fraud attempt will be detected. Detector performance depends on media quality, attack techniques, training data and the match between test conditions and real-world use. External identity databases may be unavailable or inaccessible. Some genuine users may not be able to complete camera-based checks because of device limitations or accessibility needs. Alternative verification routes and human review should therefore be available where appropriate.

A risk score is a decision-support signal rather than proof of wrongdoing. The platform must communicate uncertainty, avoid unnecessary retention of sensitive data and provide a process for users to retry, challenge or appeal a decision under the organization’s policy.

8.7 Requirement Traceability

Requirement traceability connects each requirement to its design component and verification test. For example, FR-02 (document upload) maps to the client upload interface and document-processing service and is verified by AT-01 and AT-02. FR-05 (liveness detection) maps to the biometric service and is verified with genuine and presentation-attack test samples. FR-09 (decision management) maps to the risk engine and decision workflow and is verified by AT-04, AT-05 and AT-06. FR-12 (audit logging) maps to the audit service and is verified by AT-09. This mapping helps ensure that requirements are implemented and tested before release.

9. Conclusion

The Deepfake-Aware Digital Identity Verification Platform provides a structured approach to addressing identity fraud in remote digital services. It combines document verification, biometric matching, liveness detection, media-manipulation analysis and behavioural signals within a risk-based architecture.

The SRS defines functional and non-functional requirements, actors, use cases, architecture, data relationships, interfaces, assumptions, constraints and validation procedures. It also incorporates secure APIs, audit logging, monitoring, CI/CD, privacy controls and human review.

By combining technical safeguards with privacy protection, fairness evaluation and explainable decision-making, the platform can strengthen identity verification for banking, fintech, insurance and other online services. Its real-world effectiveness depends on rigorous testing, representative data, appropriate risk thresholds and continuous security monitoring.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25